

## NORNE OIL PLATFORM, Norway

WMO: 010876

Lat: 66.017N

Lon: 8.083E

Elev: 90

StdP: 100.25

Time Zone: 1.00 (EUC)

Period: 11-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 3             | -2.1       | -1.2 | -9.2                          | 1.7 | 0.1  | -8.1 | 1.9 | 0.6  | 28.2                  | 4.5  | 26.0 | 5.2  | 12.0                  | 90   | 1.108 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 1.9                    | 16.2            | 14.3 | 15.2 | 13.6 | 14.2 | 12.8 | 14.7                | 16.2 | 14.0 | 15.3 | 13.2 | 14.4 | 7.2                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |     |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme Max WB |
|---------------------------------|------|------|------|-----|------|------|-----|------|---------------|------|------|------|------|------|----------------|
| 0.4%                            |      |      | 1%   |     |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                |
| DP                              | HR   | MCDB | DP   | HR  | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                |
| (a)                             | (b)  | (c)  | (d)  | (e) | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)            |
| 14.0                            | 10.1 | 15.5 | 13.1 | 9.5 | 14.6 | 12.8 | 9.3 | 14.3 | 41.5          | 15.8 | 39.5 | 15.2 | 37.3 | 14.2 | 17.1           |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |     |
| (4) 23.3          | 21.1 | 19.3 | DB                         | -4.2 | 18.3               | 1.0 | 1.9                                                | -4.9 | 19.7       | -5.5 | 20.9       | -6.0 | 21.9       | -6.7 | 23.4 | (4) |
| (5)               |      |      | WB                         | -5.1 | 15.3               | 0.8 | 1.4                                                | -5.6 | 16.3       | -6.1 | 17.2       | -6.5 | 18.0       | -7.1 | 19.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 7.0         | 3.4   | 3.7   | 3.2   | 4.6   | 6.8   | 9.0   | 11.7  | 12.3  | 11.0  | 8.0   | 5.8   | 4.4   | (6)  |  |
|                                                                        | DBStd       | 3.83        | 2.38  | 2.12  | 2.48  | 2.32  | 2.10  | 1.81  | 2.01  | 1.61  | 1.61  | 2.06  | 2.27  | 2.04  | (7)  |  |
|                                                                        | HDD10.0     | 1272        | 205   | 176   | 210   | 161   | 102   | 41    | 4     | 1     | 8     | 65    | 127   | 173   | (8)  |  |
|                                                                        | HDD18.3     | 4129        | 463   | 409   | 468   | 411   | 358   | 279   | 207   | 187   | 219   | 319   | 377   | 432   | (9)  |  |
|                                                                        | CDD10.0     | 183         | 0     | 0     | 0     | 0     | 2     | 11    | 55    | 72    | 39    | 4     | 0     | 0     | (10) |  |
|                                                                        | CDD18.3     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 10.1        | 12.0  | 12.0  | 11.8  | 9.4   | 8.5   | 8.3   | 7.5   | 8.1   | 9.2   | 10.7  | 11.1  | 12.4  | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1003        | 128   | 96    | 81    | 55    | 42    | 49    | 62    | 79    | 94    | 100   | 101   | 115   | (15) |  |
|                                                                        | PrecMax     | 1240        | 241   | 169   | 137   | 99    | 59    | 87    | 91    | 125   | 152   | 187   | 156   | 211   | (16) |  |
|                                                                        | PrecMin     | 605         | 70    | 29    | 23    | 27    | 22    | 21    | 21    | 38    | 36    | 34    | 23    | 39    | (17) |  |
|                                                                        | PrecStd     | 149         | 45    | 39    | 32    | 19    | 10    | 16    | 17    | 25    | 31    | 43    | 37    | 43    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 8.2   | 8.2   | 8.2   | 10.9  | 13.9  | 14.2  | 18.2  | 17.1  | 15.0  | 12.8  | 10.2  | 9.2   | (19) |  |
|                                                                        |             | MCWB        | 7.5   | 7.4   | 6.6   | 7.6   | 11.7  | 12.0  | 15.7  | 14.8  | 13.0  | 10.9  | 9.1   | 8.3   | (20) |  |
|                                                                        | 2%          | DB          | 8.0   | 7.9   | 7.9   | 9.8   | 11.1  | 13.1  | 16.9  | 16.1  | 14.1  | 11.9  | 10.0  | 8.2   | (21) |  |
|                                                                        |             | MCWB        | 7.3   | 7.1   | 6.8   | 7.7   | 9.4   | 11.3  | 15.2  | 14.2  | 12.4  | 10.0  | 8.9   | 7.4   | (22) |  |
|                                                                        | 5%          | DB          | 7.2   | 7.2   | 7.1   | 8.1   | 10.1  | 12.1  | 15.8  | 15.2  | 13.8  | 11.1  | 9.2   | 8.1   | (23) |  |
|                                                                        |             | MCWB        | 6.2   | 6.0   | 6.0   | 7.1   | 8.7   | 10.7  | 14.2  | 13.6  | 12.2  | 9.6   | 8.3   | 7.2   | (24) |  |
|                                                                        | 10%         | DB          | 6.9   | 7.0   | 6.9   | 7.2   | 9.1   | 11.8  | 14.8  | 14.9  | 13.1  | 10.8  | 8.9   | 7.2   | (25) |  |
|                                                                        |             | MCWB        | 5.8   | 5.7   | 5.8   | 6.4   | 8.2   | 10.4  | 13.3  | 13.4  | 11.7  | 9.4   | 8.0   | 6.0   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 8.0   | 7.7   | 7.5   | 8.2   | 11.6  | 12.7  | 16.3  | 15.3  | 13.3  | 11.2  | 9.8   | 8.7   | (27) |  |
|                                                                        |             | MCDB        | 8.1   | 7.8   | 7.6   | 9.1   | 14.0  | 14.2  | 18.1  | 16.8  | 14.4  | 12.3  | 10.1  | 8.7   | (28) |  |
|                                                                        | 2%          | WB          | 7.2   | 6.9   | 6.8   | 7.6   | 9.9   | 11.7  | 15.2  | 14.7  | 12.9  | 10.6  | 9.1   | 8.0   | (29) |  |
|                                                                        |             | MCDB        | 7.6   | 7.4   | 7.2   | 8.5   | 10.8  | 12.9  | 16.8  | 16.0  | 14.0  | 11.5  | 9.4   | 8.2   | (30) |  |
|                                                                        | 5%          | WB          | 6.5   | 6.2   | 6.2   | 7.1   | 8.8   | 11.1  | 14.1  | 14.1  | 12.6  | 10.1  | 8.5   | 7.3   | (31) |  |
|                                                                        |             | MCDB        | 7.1   | 7.0   | 7.0   | 7.7   | 9.5   | 12.2  | 15.4  | 15.4  | 13.7  | 10.9  | 9.0   | 7.8   | (32) |  |
|                                                                        | 10%         | WB          | 5.6   | 5.5   | 5.5   | 6.6   | 8.1   | 10.6  | 13.3  | 13.4  | 12.2  | 9.5   | 7.8   | 6.1   | (33) |  |
|                                                                        |             | MCDB        | 6.6   | 6.5   | 6.6   | 7.2   | 8.8   | 11.6  | 14.5  | 14.6  | 13.3  | 10.5  | 8.5   | 7.1   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 2.8   | 2.8   | 3.0   | 2.4   | 2.1   | 1.9   | 1.9   | 1.9   | 1.8   | 2.1   | 2.6   | 2.9   | (35) |  |
|                                                                        |             | MCDBR       | 3.2   | 2.9   | 3.0   | 2.5   | 3.3   | 3.1   | 3.7   | 2.7   | 2.3   | 2.0   | 2.4   | 3.5   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 4.0   | 3.4   | 3.6   | 2.2   | 2.4   | 2.2   | 2.5   | 2.1   | 2.3   | 2.5   | 2.8   | 4.0   | (37) |  |
|                                                                        |             | MCWBR       | 3.2   | 3.1   | 3.0   | 2.5   | 2.8   | 3.0   | 3.4   | 2.6   | 2.1   | 2.0   | 2.6   | 3.5   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.376 | 0.322 | 0.340 | 0.357 | 0.350 | 0.346 | 0.354 | 0.349 | 0.336 | 0.317 | 0.359 | 0.373 | (40) |  |
|                                                                        |             | taud        | 2.392 | 2.170 | 2.297 | 2.356 | 2.455 | 2.504 | 2.502 | 2.509 | 2.495 | 2.279 | 2.434 | 2.367 | (41) |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 171   | 516   | 718   | 805   | 855   | 867   | 846   | 814   | 735   | 523   | 176   | 27    | (42) |  |
|                                                                        |             | Edh at Noon | 21    | 55    | 78    | 94    | 94    | 91    | 89    | 80    | 65    | 48    | 19    | 8     | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 0.06  | 0.41  | 1.51  | 3.21  | 4.75  | 5.07  | 4.52  | 3.43  | 2.02  | 0.74  | 0.13  | 0.01  | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.01  | 0.05  | 0.15  | 0.29  | 0.27  | 0.56  | 0.48  | 0.29  | 0.21  | 0.12  | 0.01  | 0.00  | (45) |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| Regional (5 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page